Math 110 Midterm 2: Review of Ch. 8–9 and Drills

Notes Ch. 8–13

Section 1 covers the Ch. 8–9 material on products, quotients and the isomorphism theorems. Sections 2–8 are 48 drill questions in the format of Hammond's practice midterm, by chapter, and 24 true/false items. Each question is worth 8 or 10 points, with points per part in parentheses; at exam pace a 10-point question takes about 12 minutes. You may quote results from the notes unless a part asks you to prove them. The exam writes V^*, T^* for the dual space and dual map; the notes write V', T'. Solutions are in a separate file.

§1 Review: products, quotients and the isomorphism theorems

§1.1 Products of vector spaces

The idea. A product puts several spaces side by side. An element of $V_1 \times \cdots \times V_m$ is a list $(v_1, \ldots, v_m)$ with one entry from each space, and you add and scale entry by entry. $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ is the basic example.

Proposition (Prop. 34 / LADR 3.87, 3.89)
For vector spaces $V_1, \ldots, V_m$, the product $V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) : v_i \in V_i\}$ is a vector space with

$$(v_1, \ldots, v_m) + (w_1, \ldots, w_m) = (v_1 + w_1, \ldots, v_m + w_m), \qquad \lambda(v_1, \ldots, v_m) = (\lambda v_1, \ldots, \lambda v_m).$$

Each axiom holds because it holds in each entry. The zero vector is $(0, \ldots, 0)$ and the negative of $(v_1, \ldots, v_m)$ is $(-v_1, \ldots, -v_m)$.

Example (Ex. 39, 40)
$\mathbb{R}^n$ is $\mathbb{R} \times \cdots \times \mathbb{R}$ (n factors). $\mathbb{R}^m \times \mathbb{R}^n$ is identified with $\mathbb{R}^{m+n}$ by $\big((x_1, \ldots, x_m), (y_1, \ldots, y_n)\big) \mapsto (x_1, \ldots, x_m, y_1, \ldots, y_n)$. An element of $\mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$ is a pair such as $\big(2 + 3x,\ (1, -1)\big)$, and $\big(2 + 3x, (1, -1)\big) + \big(x, (0, 5)\big) = \big(2 + 4x, (1, 4)\big)$.

Proposition (Prop. 35, 36 / LADR 3.76, HW4 P5)
Let $V_{(i)} = \{(0, \ldots, 0, v_i, 0, \ldots, 0) : v_i \in V_i\}$ (entry v_i in position i, zeros elsewhere); $V_{(i)}$ is a copy of V_i. Then

$$V_1 \times \cdots \times V_m = V_{(1)} \oplus \cdots \oplus V_{(m)},$$

and if each V_i is finite-dimensional, $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$.

Proof. $(v_1, \ldots, v_m) = (v_1, 0, \ldots, 0) + \cdots + (0, \ldots, 0, v_m)$, and this is the only way to write it as a sum of one vector from each $V_{(i)}$, because the entries determine the pieces. For the dimension, take two spaces: if $v_1, \ldots, v_k$ is a basis of V and $w_1, \ldots, w_l$ a basis of W, then

$$(v_1, 0), \ldots, (v_k, 0),\ (0, w_1), \ldots, (0, w_l)$$

is a basis of $V \times W$. It spans, since $(v, w) = (v, 0) + (0, w)$ and each part is a combination of the listed vectors. It is independent, since $\sum a_i(v_i, 0) + \sum b_j(0, w_j) = \big(\sum a_i v_i, \sum b_j w_j\big) = (0, 0)$ forces all $a_i = 0$ and all $b_j = 0$. So $\dim(V \times W) = k + l$; repeat for more factors. □

Because of this, the product is also called the **external direct sum**. The dimension of a product is the *sum* of the dimensions (compare $\dim \mathcal{L}(V,W) = (\dim V)(\dim W)$).

Proposition (Prop. 37 / LADR 3.93)

For subspaces $U_1, \dots, U_m$ of V, the map

$$\Gamma\colon U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m, \qquad \Gamma(u_1, \dots, u_m) = u_1 + \cdots + u_m$$

is linear and surjective, and it is injective iff the sum $U_1 + \cdots + U_m$ is direct. Consequently $\dim(U_1 + \cdots + U_m) \le \dim U_1 + \cdots + \dim U_m$, with equality iff the sum is direct.

Proof. Surjective: every element of $U_1 + \cdots + U_m$ is some $u_1 + \cdots + u_m$. Injective means $u_1 + \cdots + u_m = 0$ only when every $u_i = 0$, which is the test for a direct sum. For the dimensions, rank–nullity gives $\dim(U_1 + \cdots + U_m) = \dim(U_1 \times \cdots \times U_m) - \dim \operatorname{null} \Gamma$, and $\dim(U_1 \times \cdots \times U_m) = \sum \dim U_i$. □

For two subspaces, $\operatorname{null} \Gamma = \{(u, -u) : u \in U \cap W\}$, a copy of $U \cap W$. Rank–nullity then gives the formula $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Example

In $\mathbb{R}^3$ let $U = \operatorname{span}(e_1, e_2)$ and $W = \operatorname{span}(e_2, e_3)$. $\Gamma\colon U \times W \to \mathbb{R}^3$ is surjective, and $\operatorname{null} \Gamma = \{(te_2, -te_2) : t \in \mathbb{R}\}$ has dimension 1. So $\dim(U + W) = 4 - 1 = 3$, and the sum is not direct.

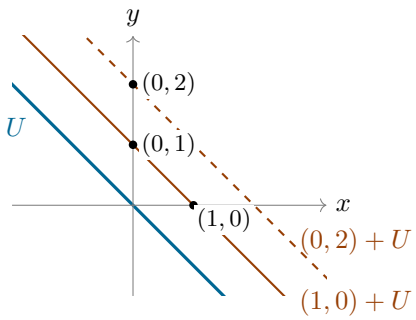
Traps.

- $V \times W$ is a new space of pairs. It is not $V \cup W$, and it is not $V + W$ (which only makes sense when V and W sit inside one larger space).
- $\dim(V \times W) = \dim V + \dim W$, not $(\dim V)(\dim W)$.

§1.2 Affine subsets and quotient spaces

The idea. The solutions of $Tx = b$ with $b \neq 0$ do not form a subspace (they do not contain 0), but they form a subspace shifted away from 0. Such shifted subspaces are called affine subsets. If we fix one subspace U and collect all of its shifts, these shifts can themselves be added and scaled: that is the quotient space V/U. In V/U we stop distinguishing two vectors whose difference lies in U.

Definition (Def. 22, 23 / LADR 3.95, 3.97)**.** For a subspace $U \subseteq V$ and $v \in V$, the **translate** $v + U = \{v + u : u \in U\}$ is an **affine subset** of V, and it is **parallel** to U. Two affine subsets are parallel if they are parallel to the same subspace.



$U = \operatorname{span}\big((-1, 1)\big)$ is the line $x + y = 0$. The line $x + y = 1$ is the affine subset $(1, 0) + U$ (Ex. 41); it is also $(0, 1) + U$, because $(1, 0) - (0, 1) = (1, -1) \in U$. The dashed line $x + y = 2$ is $(0, 2) + U$. Each of these lines is one element of the quotient space $\mathbb{R}^2/U$.

Example (Ex. 41, 42; solution sets)

- The line $\{(x, y) : x + y = 1\}$ in $\mathbb{R}^2$ is $(1, 0) + \operatorname{span}\big((-1, 1)\big)$.
- Indefinite integrals: $\int x^2\,dx = \frac{1}{3}x^3 + C$ is the affine subset $\frac{1}{3}x^3 + \mathbb{R}$ of $\mathcal{P}(\mathbb{R})$, parallel to the subspace $\mathbb{R}$ of constant polynomials.
- If $Tx_0 = b$, then $\{x : Tx = b\} = x_0 + \operatorname{null} T$: indeed $Tx = b$ iff $T(x - x_0) = 0$ iff $x - x_0 \in \operatorname{null} T$.

Proposition (Prop. 38)

An affine subset is parallel to only one subspace: if $v + U = x + W$, then $U = W$.

Proof. Since $v \in x + W$, write $v = x + w_0$ with $w_0 \in W$. For $u \in U$, $v + u \in x + W$, so $v + u = x + w$ for some $w \in W$, and $u = w - w_0 \in W$. So $U \subseteq W$; the same argument with the roles switched gives $W \subseteq U$. □

Proposition (Prop. 39 / LADR 3.101, HW4 P6)

For a subspace U and $v, w \in V$, these are equivalent:

1. $v - w \in U$;
2. $v + U = w + U$;
3. $(v + U) \cap (w + U) \neq \varnothing$.

Also, $v + U$ is a subspace iff $v \in U$, and then $v + U = U$.

Proof. (1)⇒(2): for $u \in U$, $v + u = w + \big((v - w) + u\big) \in w + U$, so $v + U \subseteq w + U$; the same argument gives the other inclusion. (2)⇒(3): $v = v + 0$ lies in both. (3)⇒(1): if $v + u_1 = w + u_2$, then $v - w = u_2 - u_1 \in U$. For the last statement: if $v + U$ is a subspace it contains 0, so $0 = v + u$ for some $u \in U$ and $v = -u \in U$. If $v \in U$, then $v - 0 \in U$, so $v + U = 0 + U = U$ by (1)⇒(2). □

So two translates are either equal or disjoint, like the parallel lines in the figure.

Definition (Def. 24)**.** The **quotient space** V/U is the set of all affine subsets of V parallel to U, with

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = (\lambda v) + U.$$

Its zero is $0 + U = U$.

Why these operations make sense ("well-defined"). The same translate has many names: $v + U = v' + U$ whenever $v - v' \in U$. The formulas above use a name v, so we must check that the answer does not depend on which name we use. If $v_1 + U = v_2 + U$ and $w_1 + U = w_2 + U$, then $v_1 - v_2 \in U$ and $w_1 - w_2 \in U$, so $(v_1 + w_1) - (v_2 + w_2) \in U$ and the two sums are the same translate. Similarly $\lambda v_1 - \lambda v_2 = \lambda(v_1 - v_2) \in U$. This kind of check is needed every time a map out of V/U is defined by a formula in v.

Example

In $\mathbb{R}^2$ with $U = \operatorname{span}(e_2)$ (the y-axis), each element of $\mathbb{R}^2/U$ is a vertical line $\{x = c\}$. $(1, 2) + U = (1, 4) + U$ because $(0, -2) \in U$: in $\mathbb{R}^2/U$ only the x-coordinate is remembered.

Proposition (Basis of a quotient)

Let $u_1, \ldots, u_m$ be a basis of U, extended to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V. Then $v_1 + U, \ldots, v_k + U$ is a basis of V/U. In particular $\dim V/U = \dim V - \dim U$.

Proof. Spanning: write $v = \sum a_i u_i + \sum b_j v_j$; then $v + U = \sum b_j(v_j + U)$, since $\sum a_i u_i \in U$. Independence: if $\sum b_j(v_j + U) = 0 + U$, then $\sum b_j v_j \in U$, so $\sum b_j v_j = \sum c_i u_i$ for some c_i. The whole list $u_1, \ldots, u_m, v_1, \ldots, v_k$ is independent, so all $b_j = 0$. □

Traps.

- An element of V/U is a set $v+U$, not a vector of V. V/U is not a subspace of V.
- $v+U=w+U$ does not mean $v=w$; it means $v-w\in U$.
- Every formula on V/U written in terms of v needs a well-definedness check.

§1.3 Quotient maps and the isomorphism theorems

The idea. Every surjective linear map $T\colon V\to W$ makes W look like a quotient of V: two vectors have the same image exactly when their difference is in $\operatorname{null} T$, so W "is" $V/\operatorname{null} T$. The isomorphism theorems make this precise, and they are the standard way to identify a quotient space with a familiar space.

Definition (Def. 25 / LADR 3.104)**.** The **quotient map** is $\pi\colon V\to V/U$, $\pi(v)=v+U$. It is linear and surjective, and $\operatorname{null}\pi=U$, since $v+U=0+U$ iff $v\in U$.

Proposition (Prop. 40 / LADR 3.105)

If V is finite-dimensional, $\dim V/U=\dim V-\dim U$.

Proof. Rank–nullity for π: $\dim V=\dim\operatorname{null}\pi+\dim\operatorname{range}\pi=\dim U+\dim V/U$. □

Theorem (Prop. 41 / LADR 3.107: first isomorphism theorem)

For $T\in\mathcal{L}(V,W)$, the map

$$\bar{T}\colon V/\operatorname{null}T\to\operatorname{range}T,\qquad \bar{T}(v+\operatorname{null}T)=Tv$$

is a well-defined isomorphism, and $T=\iota\circ\bar{T}\circ\pi$, where $\pi\colon V\to V/\operatorname{null}T$ is the quotient map and $\iota\colon\operatorname{range}T\to W$ is the inclusion $\iota(w)=w$.

$$\begin{array}{ccc} V & \xrightarrow{\;T\;} & W \\ {\scriptstyle\pi}\downarrow & & \uparrow{\scriptstyle\iota} \\ V/\operatorname{null}T & \xrightarrow{\;\bar{T}\;} & \operatorname{range}T \end{array}$$

Proof. Well defined: if $v+\operatorname{null}T=v'+\operatorname{null}T$, then $v-v'\in\operatorname{null}T$, so $Tv=Tv'$. Linear: from linearity of T. Injective: $\bar{T}(v+\operatorname{null}T)=0$ means $Tv=0$, so $v\in\operatorname{null}T$ and $v+\operatorname{null}T$ is the zero of $V/\operatorname{null}T$. Surjective: every element of $\operatorname{range}T$ is some $Tv=\bar{T}(v+\operatorname{null}T)$. □

How to use it. To show $V/U\cong W$, find a linear map $T\colon V\to W$ that is surjective and has $\operatorname{null}T=U$ exactly. Then $v+U\mapsto Tv$ is the isomorphism.

Example (Ex. 43)

Differentiation $D\colon\mathcal{P}(\mathbb{R})\to\mathcal{P}(\mathbb{R})$ is surjective with $\operatorname{null}D=\mathbb{R}$ (the constants). So $\mathcal{P}(\mathbb{R})/\mathbb{R}\cong\mathcal{P}(\mathbb{R})$ by $p+\mathbb{R}\mapsto p'$; the inverse sends p to its indefinite integral $\int p\,dx$, an element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$.

Example (A quotient of $\mathcal{P}_3(\mathbb{R})$)

Let $T\colon\mathcal{P}_3(\mathbb{R})\to\mathbb{R}^2$, $T(p)=\big(p(1),p'(1)\big)$. It is surjective: $T(1)=(1,0)$ and $T(x-1)=(0,1)$. So $\dim\operatorname{null}T=4-2=2$, and $\operatorname{null}T=\{p:p(1)=p'(1)=0\}=\operatorname{span}\big((x-1)^2,\ x(x-1)^2\big)$ (both are in it, and they are independent). By the first isomorphism theorem, $\mathcal{P}_3(\mathbb{R})/\operatorname{null}T\cong\mathbb{R}^2$ via $p+\operatorname{null}T\mapsto\big(p(1),p'(1)\big)$. A basis of the quotient is $1+\operatorname{null}T$, $(x-1)+\operatorname{null}T$, the preimages of $(1,0)$ and $(0,1)$.

Proposition (Prop. 42; HW5 P2: induced maps)
Let $T \in \mathcal{L}(V, W)$, and let $U_1 \subseteq V$, $U_2 \subseteq W$ be subspaces with $T(U_1) \subseteq U_2$. Then

$$\bar{T}\colon V/U_1 \to W/U_2, \qquad \bar{T}(v + U_1) = Tv + U_2$$

is a well-defined linear map. In particular (take $U_2 = \{0\}$): if $U \subseteq \operatorname{null} T$, there is exactly one linear $\tilde{T}\colon V/U \to W$ with $\tilde{T}(v + U) = Tv$, that is $T = \tilde{T} \circ \pi$, and $\operatorname{null} \tilde{T} = \{v + U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

Proof. Well defined: if $v + U_1 = v' + U_1$, then $v - v' \in U_1$, so $Tv - Tv' = T(v - v') \in U_2$ and $Tv + U_2 = Tv' + U_2$. Linearity comes from T. For $\tilde{T}$: it is unique because its value on every $v + U$ is forced to be Tv. And $\tilde{T}(v + U) = 0$ iff $Tv = 0$ iff $v \in \operatorname{null} T$. □

The quotient operator T/U (invariant subspaces section of the packet) is the case $V = W$ and $U_1 = U_2 = U$.

Theorem (Prop. 43: second isomorphism theorem)
For subspaces U, W of V: $U/(U \cap W) \cong (U + W)/W$, via $u + (U \cap W) \mapsto u + W$. In finite dimensions this gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.

Proof. The map $S\colon U \to (U + W)/W$, $S(u) = u + W$, is linear. It is surjective: an element $u + w + W$ of $(U + W)/W$ equals $u + W$, since $(u + w) - u \in W$. Its null space: $u + W = 0 + W$ iff $u \in W$, so $\operatorname{null} S = U \cap W$. Apply the first isomorphism theorem to S. □

Theorem (Prop. 44: third isomorphism theorem)
For subspaces $W \subseteq U \subseteq V$: $(V/W)/(U/W) \cong V/U$, via $(v+W)+U/W \mapsto v+U$. (Here $U/W = \{u+W : u \in U\}$, a subspace of V/W.)

Proof. The identity map of V sends W into U, so by Prop. 42 it induces $S\colon V/W \to V/U$, $S(v + W) = v + U$. S is surjective, and $S(v + W) = 0 + U$ iff $v \in U$ iff $v + W \in U/W$. So $\operatorname{null} S = U/W$; apply the first isomorphism theorem. □

These are used in proofs by induction on dimension: if $W \neq \{0\}$, then V/W is smaller than V, and the third isomorphism theorem relates quotients of V/W to quotients of V.

Traps.

- The first isomorphism theorem gives $V/\operatorname{null} T \cong \operatorname{range} T$, which is W only when T is surjective.
- To prove $V/U \cong W$ with it, you need $\operatorname{null} T = U$ exactly, not just $U \subseteq \operatorname{null} T$.
- A map out of a quotient defined by "$v + U \mapsto$ (formula in v)" is only a map after the well-definedness check.

§2 Products, affine subsets and quotient spaces (Ch. 8)

1. A product space. Let $V = \mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$.

a) (2) Give a basis of V and its dimension.
b) (3) Define $T\colon V \to \mathcal{P}_3(\mathbb{R})$ by $T\big(p, (a, b)\big) = p + ax^2 + bx^3$. Prove that T is an isomorphism.
c) (3) Find the dimension of $\{(p, (a, b)) \in V : p(1) = a + b\}$.

2. The sum map. Let $U_1, \ldots, U_m$ be subspaces of V and $\Gamma\colon U_1 \times \cdots \times U_m \to V$, $\Gamma(u_1, \ldots, u_m) = u_1 + \cdots + u_m$.

a) (2) Prove that Γ is linear.
b) (2) Prove that $\operatorname{range} \Gamma = U_1 + \cdots + U_m$.
c) (3) Prove that Γ is injective if and only if the sum $U_1 + \cdots + U_m$ is direct.
d) (3) Assume V is finite-dimensional. Deduce $\dim(U_1 + \cdots + U_m) \leq \dim U_1 + \cdots + \dim U_m$, with equality if and only if the sum is direct.

3. Affine subsets. Let U be a subspace of V and $v, w \in V$.

a) (3) Prove that if $v - w \in U$, then $v + U = w + U$.
b) (3) Prove that if $(v + U) \cap (w + U) \neq \varnothing$, then $v - w \in U$.
c) (2) Prove that if $v + U = x + W$ for a vector x and a subspace W, then $U = W$.
d) (2) Let $A = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 3\}$. Find the subspace U to which A is parallel and a vector v with $A = v + U$.

4. Solution sets. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$, $T(x, y, z) = (x + y + z,\ x - z)$.

a) (2) Find $\operatorname{null} T$.
b) (3) Write $\{u \in \mathbb{R}^3 : Tu = (2, 1)\}$ in the form $v + \operatorname{null} T$.
c) (3) Prove in general: for $T \in \mathcal{L}(V, W)$ and $b \in W$, the set $\{u : Tu = b\}$ is either empty or equal to $u_0 + \operatorname{null} T$ for any u_0 with $Tu_0 = b$.

5. Operations on V/U. Let U be a subspace of V.

a) (3) Prove that $(v + U) + (w + U) = (v + w) + U$ is well defined: if $v + U = v' + U$ and $w + U = w' + U$, the two answers agree.
b) (2) Prove that $\lambda(v + U) = \lambda v + U$ is well defined.
c) (3) Let W be a subspace with $U \subseteq W \subseteq V$. Prove that $W/U = \{w + U : w \in W\}$ is a subspace of V/U, and if V is finite-dimensional, that $\dim W/U = \dim W - \dim U$.
d) (2) In $\mathbb{R}^3/U$ with $U = \operatorname{span}(e_3)$, decide whether $(1, 2, 5) + U = (1, 2, -1) + U$ and whether $(1, 2, 5) + U = (1, 3, 5) + U$.

6. A basis of a quotient. Let $V = \mathbb{R}^4$ and $U = \operatorname{span}\big((1, 1, 0, 0), (0, 0, 1, 1)\big)$.

a) (3) Find a basis of V/U.
b) (3) Write $(3, 1, 4, 1) + U$ in your basis.
c) (2) Find a linear $T\colon \mathbb{R}^4 \to \mathbb{R}^2$ with $\operatorname{null} T = U$, and use it to write down an isomorphism $V/U \to \mathbb{R}^2$.

7. Quotients of an infinite-dimensional space. Let $V = \mathcal{P}(\mathbb{R})$.

a) (3) Let $U = \{p : p(0) = 0\}$. Prove V/U is 1-dimensional and give a basis.
b) (3) Let $W = \{p : p(0) = p'(0) = 0\}$. Find $\dim V/W$.
c) (2) Prove $V/\{0\} \cong V$.

§3 Quotient maps and the isomorphism theorems (Ch. 9)

8. The first isomorphism theorem. Let $T \in \mathcal{L}(V, W)$.

a) (2) Prove that $\bar{T}(v + \operatorname{null} T) = Tv$ is well defined on $V/\operatorname{null} T$.
b) (2) Prove that $\bar{T}$ is linear.
c) (3) Prove that $\bar{T}$ is injective.
d) (3) Prove that $\bar{T}$ maps onto $\operatorname{range} T$. Apply this to differentiation on $\mathcal{P}(\mathbb{R})$ to show $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$, where $\mathbb{R}$ is the subspace of constant polynomials.

9. Symmetric parts. Let $V = \mathbb{R}^{2,2}$ and $T(A) = A + A^T$.

a) (3) Find $\operatorname{null} T$ and $\operatorname{range} T$.
b) (3) Write down the isomorphism $V/\operatorname{null} T \to \operatorname{range} T$ given by the first isomorphism theorem.
c) (2) Find a basis of $V/\operatorname{null} T$ and its image.
d) (2) Check that the dimensions agree.

10. Taylor coefficients. Let $V = \mathcal{P}_4(\mathbb{R})$, $T(p) = \big(p(2), p'(2), p''(2)\big)$, and $U = \operatorname{null} T$.

a) (3) Prove $T\colon V \to \mathbb{R}^3$ is surjective and $U = \{(x - 2)^3 q : q \in \mathcal{P}_1(\mathbb{R})\}$.
b) (2) Write down the isomorphism $V/U \to \mathbb{R}^3$.
c) (3) Prove that $1 + U,\ (x - 2) + U,\ (x - 2)^2 + U$ is a basis of V/U, and find their images.
d) (2) Write $x^4 + U$ in this basis.

11. Maps out of a quotient. Let $T \in \mathcal{L}(V,W)$ and let U be a subspace with $U \subseteq \operatorname{null} T$.

a) (3) Prove there is exactly one linear $\tilde{T}\colon V/U \to W$ with $\tilde{T}(v+U) = Tv$ for all v.
b) (3) Prove $\operatorname{null}\tilde{T} = \{v+U : v \in \operatorname{null} T\}$.
c) (2) Prove $\tilde{T}$ is injective if and only if $U = \operatorname{null} T$.
d) (2) For $T(x,y,z) = x - y$ on $\mathbb{R}^3$ and $U = \operatorname{span}(e_3)$, find $\operatorname{null}\tilde{T}$. Is $\tilde{T}$ injective?

12. The second isomorphism theorem. Let U, W be subspaces of V.

a) (3) Prove that $S\colon U \to (U+W)/W$, $S(u) = u + W$, is linear and surjective.
b) (3) Prove $\operatorname{null} S = U \cap W$.
c) (2) Deduce $U/(U\cap W) \cong (U+W)/W$, and, for V finite-dimensional, $\dim(U+W) = \dim U + \dim W - \dim(U\cap W)$.
d) (2) For $U = \operatorname{span}(e_1, e_2)$ and $W = \operatorname{span}(e_2 + e_3)$ in $\mathbb{R}^3$, find $U \cap W$ and $U + W$ and check both sides.

13. The third isomorphism theorem. Let $W \subseteq U \subseteq V$ be subspaces.

a) (2) Prove that U/W is a subspace of V/W.
b) (3) Prove that $S\colon V/W \to V/U$, $S(v+W) = v + U$, is well defined, linear and surjective.
c) (3) Prove $\operatorname{null} S = U/W$ and deduce $(V/W)/(U/W) \cong V/U$.
d) (2) For $V = \mathbb{R}^3$, $W = \operatorname{span}(e_1)$, $U = \operatorname{span}(e_1, e_2)$, identify $(V/W)/(U/W)$ with $\mathbb{R}$ by an explicit formula.

14. A quotient operator. Let $T \in \mathcal{L}(\mathbb{R}^3)$ have matrix $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{smallmatrix}\right)$ in the standard basis, and $U = \operatorname{span}(e_3)$.

a) (2) Prove U is T-invariant.
b) (3) Find the matrix of T/U in the basis $e_1 + U,\ e_2 + U$.
c) (3) Find the eigenvalues of $T|_U$, T/U and T. Is T/U diagonalizable?
d) (2) Without computing eigenvectors of T, decide whether T is diagonalizable.

§4 Duality (Ch. 10)

15. Dual bases. Let $v_1, \dots, v_n$ be a basis of V with dual basis $\varphi_1, \dots, \varphi_n$.

a) (3) Prove $\varphi_1, \dots, \varphi_n$ is linearly independent.
b) (2) Deduce that it is a basis of V'.
c) (3) Prove $\psi = \sum_j \psi(v_j)\varphi_j$ for every $\psi \in V'$.
d) (2) Find the dual basis of $(1,2), (3,4)$ in $\mathbb{R}^2$.

16. Functionals on $\mathcal{P}_2(\mathbb{R})$. Use the basis $1,\ x-1,\ (x-1)^2$ of $V = \mathcal{P}_2(\mathbb{R})$.

a) (3) Prove that the dual basis is $\varphi_1(p) = p(1)$, $\varphi_2(p) = p'(1)$, $\varphi_3(p) = \frac{1}{2}p''(1)$.
b) (3) Write $\psi(p) = \int_0^1 p(x)\,dx$ in this dual basis.
c) (2) Write $\varepsilon(p) = p(0)$ in this dual basis.
d) (2) Is $\varphi_1, \psi, \varepsilon$ a basis of V'?

17. The double dual. For $v \in V$ define $j_V(v)\colon V' \to \mathbb{F}$ by $j_V(v)(\varphi) = \varphi(v)$.

a) (2) Prove $j_V(v)$ is linear, so $j_V(v) \in V''$.
b) (2) Prove $j_V\colon V \to V''$ is linear.
c) (3) Prove j_V is injective.
d) (1) Deduce that j_V is an isomorphism when V is finite-dimensional.

18. The dual of differentiation. Let $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_1(\mathbb{R})$, $Tp = p'$. Let $\varphi_0, \varphi_1, \varphi_2$ be the dual basis of $1, x, x^2$ and ψ_0, ψ_1 the dual basis of $1, x$.

a) (3) Find $\mathcal{M}(T)$ and $\mathcal{M}(T')$.
b) (3) Compute $T'(\psi_0)$ and $T'(\psi_1)$ directly from the definition of the dual map.
c) (2) Find $\operatorname{null} T'$ and $\operatorname{range} T'$, and check them against $\operatorname{null} T' = (\operatorname{range} T)^0$ and $\operatorname{range} T' = (\operatorname{null} T)^0$.
d) (2) Let $\varepsilon(q) = q(1)$ on $\mathcal{P}_1(\mathbb{R})$. Write $T'(\varepsilon)$ in terms of the φ_j.

19. Properties of the dual map. Let S, T be linear maps.

a) (2) Prove $(S+T)' = S' + T'$ and $(cT)' = cT'$ for $S, T \in \mathcal{L}(V, W)$.
b) (3) Prove $(ST)' = T'S'$ for $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$.
c) (3) Prove that if $T \in \mathcal{L}(V, W)$ is invertible, then T' is invertible and $(T')^{-1} = (T^{-1})'$.
d) (2) Find $\mathcal{M}(T')$ in the dual bases for $T(x, y) = (x + y,\ 2x,\ 3y)$, $T\colon \mathbb{R}^2 \to \mathbb{R}^3$.

20. Computing annihilators.

a) (3) Find a basis of U^0 for $U = \operatorname{span}\big((1,0,1,0),(0,1,0,1)\big) \subseteq \mathbb{R}^4$ (functionals as row vectors).
b) (3) In $\mathcal{P}_3(\mathbb{R})$ let $U = \{p : p(1) = p(2) = 0\}$ and $\varepsilon_a(p) = p(a)$. Prove $U^0 = \operatorname{span}(\varepsilon_1, \varepsilon_2)$.
c) (2) Let $W = \mathcal{P}_1(\mathbb{R}) \subseteq \mathcal{P}_3(\mathbb{R})$. Is $\varepsilon_1 - \varepsilon_2 \in W^0$?
d) (2) Find $\dim W^0$ and one nonzero element of W^0.

21. Null space and range of T'. Let $T \in \mathcal{L}(V, W)$ with V, W finite-dimensional.

a) (3) Prove $\operatorname{null} T' = (\operatorname{range} T)^0$.
b) (2) Prove $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$.
c) (3) Prove $\operatorname{range} T' = (\operatorname{null} T)^0$ by comparing dimensions.
d) (2) Deduce that T is surjective iff T' is injective, and that a matrix and its transpose have the same rank.

22. Annihilators in general. Let V be finite-dimensional and U, W subspaces.

a) (3) Prove that the restriction map $V' \to U'$, $\varphi \mapsto \varphi|_U$, is surjective with null space U^0, and deduce $\dim U^0 = \dim V - \dim U$.
b) (3) Prove $U \subseteq W$ if and only if $W^0 \subseteq U^0$.
c) (2) Prove $(U + W)^0 = U^0 \cap W^0$.
d) (2) Prove: if $\varphi_1, \dots, \varphi_m \in V'$ and $\operatorname{null}\varphi_1 \cap \cdots \cap \operatorname{null}\varphi_m = \{0\}$, then $m \geq \dim V$.

23. When functionals form a basis. Let $\dim V = n$ and $\varphi_1, \dots, \varphi_n \in V'$; define $S\colon V \to \mathbb{F}^n$, $S(v) = \big(\varphi_1(v), \dots, \varphi_n(v)\big)$.

a) (3) Prove that $\varphi_1, \dots, \varphi_n$ is linearly independent iff S is surjective. (Hint: S' sends the ith coordinate functional on $\mathbb{F}^n$ to φ_i.)
b) (3) Deduce that $\varphi_1, \dots, \varphi_n$ is a basis of V' iff the only v with $\varphi_i(v) = 0$ for all i is $v = 0$.
c) (2) Decide whether $p \mapsto p(0)$, $p \mapsto p(1)$, $p \mapsto p(2)$ is a basis of $\mathcal{P}_2(\mathbb{R})'$.

§5 Invariant subspaces and eigenvalues (Ch. 11)

24. Invariant subspaces of a 3×3 matrix. Let $T(x, y, z) = (2x,\ x + 2y,\ 5z)$ on $\mathbb{R}^3$.

a) (3) Find the eigenvalues and eigenspaces of T.
b) (3) Decide which of $\operatorname{span}(e_1)$, $\operatorname{span}(e_1, e_2)$, $\operatorname{span}(e_2, e_3)$ are T-invariant.
c) (2) List all 1-dimensional invariant subspaces.
d) (2) Is T diagonalizable?

25. Existence of eigenvalues. Let $V \neq \{0\}$ be a finite-dimensional complex vector space, $n = \dim V$, and $T \in \mathcal{L}(V)$.

a) (2) Fix $v \neq 0$. Prove there is a nonconstant polynomial p of degree at most n with $p(T)v = 0$.
b) (3) Choose such p of smallest degree and factor $p = (x - \lambda)q$. Explain why this factorization exists and why $q(T)v \neq 0$.
c) (3) Prove that $w = q(T)v$ is an eigenvector of T.
d) (2) Where does this argument break down for the rotation $R(x, y) = (-y, x)$ on $\mathbb{R}^2$?

26. Independence of eigenvectors. Let $T \in \mathcal{L}(V)$.

a) (4) Prove that eigenvectors $v_1, \dots, v_m$ for distinct eigenvalues $\lambda_1, \dots, \lambda_m$ are linearly independent.
b) (2) Deduce that T has at most $\dim V$ distinct eigenvalues if V is finite-dimensional.
c) (2) Prove that e^x, e^{2x}, e^{3x} are linearly independent functions.
d) (2) Show that every $\lambda \in \mathbb{R}$ is an eigenvalue of the backward shift $(a_1, a_2, a_3, \dots) \mapsto (a_2, a_3, \dots)$ on real sequences. Why does this not contradict (b)?

27. A reflection of polynomials. Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, $(Tp)(x) = p(-x)$.

a) (2) Prove $T^2 = I$ and deduce that every eigenvalue is 1 or -1.
b) (3) Find $E(1, T)$ and $E(-1, T)$.
c) (3) Prove T is diagonalizable in two ways: by dimensions of eigenspaces, and by the minimal polynomial.
d) (2) Find the eigenvalues of T' and the dimensions of its eigenspaces.

28. Diagonalizing a 2×2 matrix. Let $A = \left(\begin{smallmatrix} 4 & -2 \\ 1 & 1 \end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of A.
b) (3) Find P invertible and D diagonal with $A = PDP^{-1}$.
c) (2) Find a formula for A^k.
d) (2) Find m_A.

29. Eigenvalues that survive changes.

a) (3) Let S be invertible and $R = S^{-1}TS$. Prove T and R have the same eigenvalues and $E(\lambda, R) = S^{-1}\big(E(\lambda, T)\big)$.
b) (3) Prove that T and T' have the same eigenvalues with the same geometric multiplicities (V finite-dimensional).
c) (2) For $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ find $E(1, A)$ and $E(1, A^T)$. Are they equal?

30. Stochastic matrices. Let P be an $n \times n$ column-stochastic matrix.

a) (3) Prove every eigenvalue λ of P satisfies $|\lambda| \le 1$.
b) (3) Prove that 1 is an eigenvalue of P. (Hint: $P^T\mathbf{1} = \mathbf{1}$ for $\mathbf{1} = (1, \dots, 1)$.)
c) (2) Find the stationary distribution of $P = \begin{pmatrix} 1/2 & 1/5 \\ 1/2 & 4/5 \end{pmatrix}$.
d) (2) Find the other eigenvalue of this P.

31. PageRank. Let P be column-stochastic, v a probability vector, $0 < \alpha < 1$.

a) (3) Prove $I - \alpha P$ is invertible.
b) (3) For $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $v = (1, 0)$, $\alpha = \frac{1}{2}$, solve $r = \alpha P r + (1 - \alpha)v$.
c) (2) Check that your r is a probability vector and explain in words why $r_1 > r_2$.

32. Eigenvalues of restriction and quotient. Let V be finite-dimensional, $T \in \mathcal{L}(V)$, U a T-invariant subspace.

a) (3) Prove every eigenvalue of $T|_U$ is an eigenvalue of T.
b) (3) Prove: if $Tv = \lambda v$ with $v \notin U$, then λ is an eigenvalue of T/U.
c) (4) Suppose λ is not an eigenvalue of T. Prove $T - \lambda I$ maps U onto U, and deduce that $(T/U) - \lambda I$ is injective. Conclude that every eigenvalue of T/U is an eigenvalue of T.

§6 Upper-triangular matrices and diagonalizability (Ch. 12)

33. Upper-triangular matrices. Let $v_1, \dots, v_n$ be a basis of V and $T \in \mathcal{L}(V)$.

a) (3) Prove: if $\mathcal{M}(T)$ is upper-triangular, then $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j.
b) (3) Prove: if $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j, then each $\operatorname{span}(v_1, \dots, v_j)$ is T-invariant.
c) (2) Prove: if each $\operatorname{span}(v_1, \dots, v_j)$ is T-invariant, then $\mathcal{M}(T)$ is upper-triangular.

34. Eigenvalues on the diagonal. Let $\mathcal{M}(T)$ be upper-triangular with diagonal $\lambda_1, \dots, \lambda_n$.

a) (4) Prove that λ is an eigenvalue of T iff $\lambda = \lambda_k$ for some k.
b) (2) Deduce that T is invertible iff no λ_k is 0.
c) (2) Let $T = I + D$ on $\mathcal{P}_2(\mathbb{R})$, with D differentiation. Find the eigenvalues and eigenspaces of T.
d) (2) Write T^{-1} as a polynomial in D.

35. Triangularizing a matrix. Let $A = \left(\begin{smallmatrix} 3 & 1 \\ -4 & -1 \end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of A.
b) (4) Find a basis in which the operator $x \mapsto Ax$ has an upper-triangular matrix, and the matrix S with $S^{-1}AS$ upper-triangular.
c) (3) Prove A is not diagonalizable and find m_A.

36. A diagonalizability test with a parameter. For $t \in \mathbb{R}$ let $A_t = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & t & 2 \end{smallmatrix}\right)$.

a) (2) Find the eigenvalues of A_t.
b) (4) Find all t for which A_t is diagonalizable.
c) (4) For those t, find an eigenbasis.

37. Conditions for diagonalizability. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(V)$, V finite-dimensional.

a) (2) Prove: $\mathcal{M}(T)$ is diagonal in a basis iff that basis consists of eigenvectors.
b) (3) Prove the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct.
c) (3) Prove V has an eigenbasis iff $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.
d) (2) Prove this holds iff $\dim V = \sum_i \dim E(\lambda_i, T)$.

38. Projections. Let $V \neq \{0\}$ be finite-dimensional and $P \in \mathcal{L}(V)$ with $P^2 = P$.

a) (3) Prove $V = \operatorname{null} P \oplus \operatorname{range} P$.
b) (2) Prove $\operatorname{range} P = E(1, P)$.
c) (3) Prove P is diagonalizable, and list the possible minimal polynomials of P.
d) (2) Find the matrix of the projection of $\mathbb{R}^2$ onto $\operatorname{span}\big((1,1)\big)$ along $\operatorname{span}\big((1,0)\big)$.

39. A recurrence. Let $a_0 = 0$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$.

a) (3) Find a 2×2 matrix A with $A\left(\begin{smallmatrix} a_n \\ a_{n+1} \end{smallmatrix}\right) = \left(\begin{smallmatrix} a_{n+1} \\ a_{n+2} \end{smallmatrix}\right)$, and its eigenvalues and eigenvectors.
b) (4) Write $\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ in the eigenbasis.
c) (3) Deduce a formula for a_n, and check it for $n = 2, 3$.

40. A two-state Markov chain. Let $P = \left(\begin{smallmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of P.
b) (3) Write $p_0 = (1, 0)$ in the eigenbasis, with the eigenvector for 1 scaled to be a probability vector.
c) (2) Find $P^k p_0$.
d) (2) Find $\lim_k P^k p_0$, and say how fast the error shrinks.

§7 The minimal polynomial (Ch. 13)

41. Minimal polynomials from the definition. Let $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ and $B = \operatorname{diag}(1, 1, 2)$.

a) (3) Find m_A directly from the definition (the first power of A that is a combination of lower powers).
b) (3) Find m_B.
c) (2) Which of A, B is diagonalizable?
d) (2) Find m_{A-I} and m_{2B}.

42. Polynomials that annihilate T. Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

a) (4) Prove that $q(T) = 0$ iff m_T divides q.
b) (3) Let V be complex and $T^4 = T$. Prove T is diagonalizable and list its possible eigenvalues.
c) (3) Give $T \in \mathcal{L}(\mathbb{R}^2)$ with $T^4 = T$ that is not diagonalizable over $\mathbb{R}$. Which real numbers can be eigenvalues of a real T with $T^4 = T$?

43. The degree bound. Let $\dim V = n$ and $T \in \mathcal{L}(V)$. Prove $\deg m_T \leq n$ by induction on n.

a) (2) For $v \neq 0$, find a nonzero q of degree at most n with $q(T)v = 0$. Handle the case $q(T) = 0$.
b) (3) If $q(T) \neq 0$, prove $U = \operatorname{null} q(T)$ is T-invariant and $\{0\} \neq U \neq V$.
c) (3) Prove m_T divides $m_{T|_U}\, m_{T/U}$.
d) (2) Finish the induction.

44. Distinct roots give diagonalizability. Let V be finite-dimensional and $T \in \mathcal{L}(V)$ with $m_T = (x-\lambda_1)\cdots(x-\lambda_m)$ with distinct $\lambda_i \in \mathbb{F}$, $m \geq 2$, and $g(x) = (x-\lambda_2)\cdots(x-\lambda_m)$.

a) (3) Show $g = (x-\lambda_1)q + s$ for a polynomial q and a nonzero constant s.
b) (3) For $v \in V$ let $u = \frac{1}{s}g(T)v$. Prove $u \in \operatorname{null}(T-\lambda_1 I)$ and $v - u \in \operatorname{range}(T-\lambda_1 I)$.
c) (2) Deduce $V = \operatorname{null}(T-\lambda_1 I) \oplus \operatorname{range}(T-\lambda_1 I)$.
d) (2) Finish the proof that T is diagonalizable by induction on $\dim V$.

45. A companion matrix. Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}$.

a) (3) Show that e_1, Ae_1, A^2e_1 is a basis, and find m_A.
b) (2) Is A diagonalizable over $\mathbb{R}$?
c) (3) Write A^{-1} as a polynomial in A.
d) (2) Write A^4 as a combination of I, A, A^2.

46. Minimal polynomial of a direct sum. Let V be finite-dimensional, $T \in \mathcal{L}(V)$, and $V = U \oplus W$ with U, W both T-invariant.

a) (3) Prove $p(T) = 0$ iff $p(T|_U) = 0$ and $p(T|_W) = 0$.
b) (3) Deduce that m_T is the monic polynomial of least degree divisible by both $m_{T|_U}$ and $m_{T|_W}$.
c) (2) Deduce that T is diagonalizable iff $T|_U$ and $T|_W$ are.
d) (2) Find m_T for the block matrix $\begin{pmatrix} 1 & 1 & & \\ 0 & 1 & & \\ & & 2 & 0 \\ & & 0 & 1 \end{pmatrix}$.

47. An operator on polynomials. Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, $(Tp)(x) = xp'(x)$.

a) (3) Find $\mathcal{M}(T)$ in the basis $1, x, x^2, x^3$ and the eigenvalues of T.
b) (3) Find m_T.
c) (2) Find m_{T^2}.
d) (2) Find a polynomial p such that $p(T)$ is the projection onto $\operatorname{span}(x^2)$ along $\operatorname{span}(1, x, x^3)$.

48. Real operators. Let $V \neq \{0\}$ be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$.

a) (4) Prove T has an invariant subspace of dimension 1 or 2. (Factor m_T over $\mathbb{R}$.)
b) (2) Prove that if $\dim V$ is odd, T has an eigenvalue. (Induct, using (a) and the quotient.)
c) (2) Let $T(x, y, z, w) = (-y, x, -w, z)$ on $\mathbb{R}^4$. Show T has no eigenvalue, and give a 2-dimensional invariant subspace.

§8 Quick true or false

Give a one-line reason or counterexample. Spaces are finite-dimensional.

1. $V \times W$ and $W \times V$ are isomorphic.
2. If $U_1 \cap U_2 = U_1 \cap U_3 = U_2 \cap U_3 = \{0\}$, then $U_1 + U_2 + U_3$ is a direct sum.
3. An affine subset $v + U$ contains 0 iff $v \in U$.
4. $V/\{0\} \cong V$ and $V/V = \{0 + V\}$.
5. If $T\colon V \to W$ is surjective, then $V/\operatorname{null} T \cong W$.
6. If $V/U \cong V/W$, then $U = W$.
7. $\dim U/(U \cap W) = \dim(U + W) - \dim W$.
8. Every linear map $V/U \to W$ has the form $v + U \mapsto Tv$ for some $T \in \mathcal{L}(V, W)$ with $U \subseteq \operatorname{null} T$.
9. If $\varphi \in V'$ is nonzero, then $\dim \operatorname{null} \varphi = \dim V - 1$.
10. If T is injective, then T' is injective.
11. $(U \cap W)^0 = U^0 \cap W^0$.
12. If $v_1, \ldots, v_n$ is a basis with dual basis $\varphi_1, \ldots, \varphi_n$, then φ_1 depends only on v_1.
13. Every operator on $\mathbb{R}^2$ has an eigenvalue.
14. If $T^2 = T$, then 0 and 1 are both eigenvalues of T.
15. The sum of two eigenvectors of T is an eigenvector of T or 0.
16. If λ is an eigenvalue of T, then λ^2 is an eigenvalue of T^2.
17. A 2×2 matrix whose characteristic polynomial has a double root is not diagonalizable.

18. If T is invertible and diagonalizable, then T^{-1} is diagonalizable.
19. If T is diagonalizable and U is T-invariant, then $T|_U$ is diagonalizable.
20. Every operator on a real space has an upper-triangular matrix in some basis.
21. $\deg m_T \leq \dim V$.
22. If $p(T) = 0$, then every root of p is an eigenvalue of T.
23. If T is invertible, then $m_T(0) \neq 0$.
24. If $m_T = (x-1)(x-2)$, then $\dim V = 2$.